

Hard Stop 3: Early Implementation Validation Package

AI-use disclosure. Drafted with Codex (GPT-5, Codex desktop app) from the W6 Canvas assignment screenshot, committed SafeExec code, and generated local/droplet validation logs. AI-drafted, student-revised. Full audit trail: `docs/ai-use-log.md`.

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Course: CISC 699-50-A-2026/Summer - Applied Project in Computer Information Science

Project: SafeExec - Hardened Python Execution Sandbox for LLM Agent Tool-Use

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Submission date: 2026-06-17

Repository: <https://github.com/liangzixuan/cisc-699>

Validation harness commit: 674fa58

1. Validation Objective

Hard Stop 3 asks whether the project is technically alive enough to support meaningful evidence collection before the midpoint review. For SafeExec, the current validation target is the deepest implemented path available in W6:

- request/result models and backend selection;
- local development backend with timeout and structured output;
- stdlib JSON POST /execute API shell;
- Docker backend using the hardened command shape;
- gVisor backend using Docker runtime runsc;
- repeatable validation workflow that emits JSON and text evidence.

This is still early validation, not final performance or security evaluation.

The results show that the technical path now runs repeatedly on the target

Ubuntu droplet, while also identifying environment issues that must be cleaned up before the midpoint demo.

2. Implemented Workflow

Layer	Implemented path	Evidence
Local service path	ExecutionRequest -> backend selection -> LocalSubprocessBackend -> ExecutionResult	evidence/validation-summary.txt, evidence-target-rerun/validation-summary.txt
API path	In-process HTTP server handles POST /execute and returns structured JSON	api-trace.json in local and target evidence directories
Docker path	DockerBackend runs python:3.11-slim with no network, non-root user, read-only rootfs, dropped capabilities, no-new-privileges, memory/CPU/PID limits, and tmpfs workspace	evidence-target-rerun/validation-summary.txt
gVisor path	Same Docker command with --runtime runsc	evidence-target-rerun/validation-summary.txt
Repeatability	Three iterations per case for local, Docker, and gVisor validation	validation-results.json and validation-summary.txt

3. Environment and Toolchain Snapshot

Environment	Key facts	Interpretation
Local authoring machine	macOS/Darwin ARM64, Python 3.14.5. Docker CLI installed but daemon unavailable.	Suitable for code authoring, unit tests, local/API validation. Not suitable for container evidence in this session.
Target validation host	Ubuntu 22.04 droplet, Linux 5.15.0-179-generic x86_64, Docker 29.5.0, runsc release-20260511.0.	Official environment for Docker/gVisor validation.
Target host caveats	Host Python is 3.10.12 and make is not installed. Container execution uses python:3.11-slim.	The backend path validates Python 3.11 execution, but the host harness should be moved to Python 3.11 or a dev container before W7.

Local Docker output confirms the local daemon limitation:

```
Cannot connect to the Docker daemon at unix:///var/run/docker.sock. Is the docker daemon running?
```

Target environment excerpt:

```
system: Linux 5.15.0-179-generic x86_64
python: 3.10.12
git_head: 674fa58f5a21b35a7121fdcf990725a761cce23
docker_client: ok Client: Docker Engine - Community
runsc: ok runsc version release-20260511.0
make: not-ready [Errno 2] No such file or directory: 'make'
```

4. Validation Cases

Case	Purpose	Expected result
hello_stdout	Basic stdout path.	Status OK, exact stdout.
deterministic_arithmetic	Deterministic computation.	Status ok, stdout 285.
stderr_and_exit_code	Error-channel and nonzero exit handling.	Status error, exit code 3, exact stderr.
timeout_control	Wall-time containment in local backend.	Status timeout, containment reason wall_timeout.
container_hello	Docker/gVisor container execution and non-root UID output.	Status ok.
network_disabled_probe	Early no-network validation for Docker/gVisor.	Status OK with connection failure output, not network-open.
api_trace	POST /execute request/response path.	HTTP 200 and stdout api validation.

5. Results

Local validation

Local tests passed:

```
Ran 7 tests in 1.356s
OK
```

Local validation workflow passed local service and API checks:

```
include_docker: False
all_passed: True
- local: 12/12 passed
- api_trace: passed
```

Target-host validation, first run

The first target-host Docker/gVisor validation run found one reproducibility

defect:

```
include_docker: True
all_passed: False
- local: 12/12 passed
- docker: 5/6 passed
- gvisor: 6/6 passed
- api_trace: passed
docker::container_hello#1 FAIL status=timeout exit=None reason=wall_timeout
```

The failure was not a program-correctness failure. It occurred because the first

Docker run spent the timeout window pulling python:3.11-slim. The image pull

completed, and subsequent Docker cases passed.

Target-host validation, clean rerun

After the image was cached, the clean target-host rerun passed:

```
include_docker: True
all_passed: True
- local: 12/12 passed
- docker: 6/6 passed
- gvisor: 6/6 passed
- api_trace: passed
```

The cached image evidence is:

```
python:3.11-slim ae52c5bef62a 188MB 47.7MB
sha256:ae52c5bef62a6bdd42cd1e8dffef86b9cd284bde9427da79839de7a4b983e7ca amd64 linux
```

6. Interpretation

The validation supports three claims:

1. The SafeExec execution contract is now repeatable. The same validation harness emits structured JSON and text summaries on both the local authoring machine and the Ubuntu target host.
2. The early Docker and gVisor paths are technically alive. Both backends ran sample Python programs under the intended command shape and passed the network-disabled probe after the base image was cached.
3. The project is ready for W7 expansion, but not for final security claims.

The current validation is smoke/integration evidence. It is not yet the full functional corpus, adversarial benchmark, or performance study promised for W8-W14.

The most important finding is the image-pull timeout. This is a practical reproducibility issue: a timed validation run should not include dependency download time unless the experiment is explicitly measuring cold setup. The W7 workflow should pre-pull and record image digests before timed tests.

7. Defects, Bottlenecks, and Workarounds

Finding	Evidence	Status	Next action
First Docker run timed out during image pull.	evidence-target/validation-summary.txt	Mitigated	Pre-pull python:3.11-slim and record digest before validation.
Local Docker daemon unavailable on macOS.	evidence/local-docker-version.txt	Accepted limitation	Use Ubuntu droplet as official

			container validation host.
Target host lacks make.	evidence-target/ environment- snapshot.txt	Open	Install make or document direct Python equivalents.
Target host Python is 3.10.12.	evidence-target/ environment- snapshot.txt	Open	Install Python 3.11 or run the harness in a Python 3.11 dev container.
Functional/adversarial corpora are still small.	Validation case list	Open	Add HumanEval/MBPP manifest and program-level adversarial cases.

8. Readiness for Midpoint Review

The project is on track for midpoint review if W7 focuses on evidence growth rather than new documentation. The immediate W7 targets are:

1. make target-host setup reproducible with make or equivalent documented commands;
2. pre-pull and pin the Python container image digest;
3. add the first functional corpus manifest and at least 30 runnable programs;
4. add initial adversarial programs for resource, network, process, filesystem, and environment probes;
5. capture a short midpoint demo showing local, Docker, and gVisor paths through the same validation harness.

9. Submitted Evidence

Evidence file	Meaning
evidence/environment-snapshot.txt	Local authoring environment and Docker-daemon limitation.
evidence/local-test-output.txt	Local unit-test output.
evidence/local-validation-output.txt	Local validation command output.
evidence/validation-summary.txt	Local/API validation summary.
evidence-target/environment-snapshot.txt	Ubuntu droplet environment snapshot.
evidence-target/target-test-output.txt	Target-host unit-test output.
evidence-target/validation-summary.txt	First Docker/gVisor run with image-pull timeout.
evidence-target/docker-image-python311.txt	Cached image ID/digest evidence.
evidence-target-rerun/validation-summary.txt	Clean repeated Docker/gVisor validation run.

evidence-target-rerun/validation-results.json	Machine-readable target-host validation results.
known-issues-risk-log.md	Updated defect/risk tracker.